

New Student Cheat Sheet

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1 Welcome to Nautilica!

This is a quick intro to where to find lab methods, code, old data, and calculations you might need during your project to calculate TTDs and anthropogenic carbon. Everything suggested here is available in the <https://github.com/titanica-nautilica> GitHub organisation. The new student onboarding repo <https://github.com/titanica-nautilica/new-student-onboarding> is public, so you can browse it before joining.

2 Getting Started with GitHub

GitHub is used for code versioning, sharing analysis scripts, and accessing onboarding materials. You will need a personal GitHub account to access the group repositories. You will then be able to access data and code even if you are not affiliated with either ETH Zurich or ICM Barcelona.

2.1 Creating a GitHub Account

1. Go to <https://github.com> and click **Sign up**.
2. Enter your email address, create a password, and choose a username. We recommend using a professional username (e.g. your name or initials) since this account will follow you beyond your time in the group.
3. Complete the verification steps and confirm your email address.

2.2 Joining the Nautilica GitHub Organisation

1. Send your GitHub username to your supervisor or the group admin so that an invitation can be sent to your account.
2. Check your email for an invitation from GitHub, or go to <https://github.com/notifications> to find it.
3. Accept the invitation. You will now have access to the repositories at <https://github.com/titanica-nautilica>.
4. To make your membership publicly visible on your profile, go to <https://github.com/orgs/titanica-nautilica/people>, find your name, and change the visibility dropdown from **Private** to **Public**.

3 Toolboxes to Make Your Life Easier - links to original sources

The following toolboxes are all available in the new student repo (good as of 2026), but links are included below.

- **TEOS-10 MATLAB Toolbox**: seawater properties (density, salinity, potential temperature) following the modern thermodynamic standard. <https://www.teos-10.org/software.htm#1>
- **m-map** (mapping + ETOPO bathymetry): mapping package for MATLAB; includes ETOPO bathymetry for ocean plots <https://www.eoas.ubc.ca/~rich/map.html>
- **cmocean**: perceptually uniform colourmaps designed for oceanographic data (e.g. thermal, haline, deep) <https://www.mathworks.com/matlabcentral/fileexchange/57773-cmocean-perceptually-uniform-colormaps>
- **Crameri scientific colormaps**: perceptually uniform and colourblind-safe; good for publication-ready figures <https://www.fabiocrameri.ch/colormaps/>

For bathymetry either of the following sources. Use the interactive tools to select the region you want, then export it as a netCDF file.

- **GEBCO global bathymetry** (15 arc-second gridded):
https://www.gebco.net/data_and_products/gridded_bathymetry_data/
- **ETOPO global relief** (NOAA; topography + bathymetry):
<https://www.ncei.noaa.gov/products/etopo-global-relief-model>

4 Your Data

4.1 Cruise Prep

Material for sample collection needs to be shipped to various locations. For shipping material out, there's a pro forma invoice and a shipping label template in the shipping repo <https://github.com/titanica-nautilica/cruise-prep>. Be aware that for shipments to some countries, a broker is required to clear materials through customs (e.g., Canada). If you're taking part in a cruise with the Canadian Coast Guard, leave plenty of time to get your background police checks completed - they require fingerprints. Other expeditions have medical checks. Make sure you take a large (10 L) cubitainer to collect a sample to be your internal standard for iodine processing.

4.2 Lab Methods

To process iodine, uranium or C14 samples, everything you need is in <https://github.com/titanica-nautilica/lab-methods>. There are step-by-step instruction sheets and separate log sheets for processing the samples.

4.3 Data Processing

Iodine samples measured at ETH are processed using the BATS software. It's a JavaScript programme, uploaded in the (which repo?); you also need a username and password to log in to the LIP network to use it. Contact Christof Vockenhuber at LIP. Uranium data is processed in Excel. If your samples are measured at LIP, you're likely to receive the final data sheets and not process it yourself.

Once you have your iodine data from BATS, you need to process it to calculate measurement uncertainties and correct your data. There's an Excel to do this, or a MATLAB script (**I129-post-BATS-calc.m**).

Finally, make sure you have the TEOS-10 toolbox. Calculate the following variables and add them to your data files. There is a template of how final data sheets should be set out, called **data_template_for_publishing.xlsx**.

- Depth from pressure: **gsw_z_from_p(p, lat)**
- Absolute Salinity: **SA = gsw_SA_from_SP(SP, p, lon, lat)**
- Conservative Temperature: **CT = gsw_CT_from_t(SA, t, p)**
- Potential Temperature: **pt = gsw_pt0_from_t(SA, t, p)**
- Potential Density Anomalies with reference pressures of 0, 1000 and 2000:
sigma0 = gsw_sigma0(SA, CT)
sigma1 = gsw_sigma1(SA, CT)
sigma2 = gsw_sigma2(SA, CT)
- In-situ density ρ : **gsw_rho(SA, CT, p)**

Once you have these, you can convert your tracer concentrations from **at kg⁻¹** to **at L⁻¹** (multiply by in-situ seawater density (ρ) you just calculated). Uranium is generally reported at $\times 10^6$ at L⁻¹, iodine is generally reported at $\times 10^7$ at L⁻¹.

This is the final data that will be published at Zenodo and MARIS. Make sure you join the TITANICA community on Zenodo and add it to the community when you publish.

Now that you have these final numbers, you can calculate TTDs!

5 The TTD

Real ocean circulation is turbulent and diffusive. A water sample collected at depth is actually a mixture of parcels that left the chosen initial point at many different times, via many different pathways. The Transit Time Distribution (TTD), written $G(x, t')$ and also referred to as the Green's function, describes this mixture — it is the probability distribution of transit times for water reaching location x . Rather than a single age, each water parcel has an age spectrum.

The shape of the TTD is characterised by three quantities:

- Γ (mean transit time) — the first moment of the distribution; the average time since the water was last at the surface.
- Δ (width) — controls how spread out the age spectrum is. The ratio Δ/Γ describes the relative importance of mixing vs. advection. Typical open-ocean values are near 1 (Waugh et al., 2004), but in the Arctic may reach 1.8 (Stöven et al., 2015). The Péclet number is related by $Pe = (\Gamma/\Delta)^2$ (Hall et al., 2002).
- t_{mode} (most probable transit time) — introduced by Smith et al. (2011), this is the peak of the distribution: the single transit time that is most represented in the water sample. Because the inverse-Gaussian is right-skewed, t_{mode} is always less than Γ .

5.1 How the calculation works

The TTD method is based on the following integral, which represents the sampled concentration c of a tracer at time t and location x :

$$c(x, t) = \int_0^\infty c_0(t - t') e^{-\lambda t'} G(x, t') dt'. \quad (1)$$

Here $c_0(t)$ is the tracer input function, and $G(x, t')$ is the unknown Green's function (the age spectrum). λ is the radioactive decay constant and only affects ^{137}Cs ; for ^{236}U and ^{129}I , $e^{-\lambda t'} \approx 1$. Because G has two free parameters (Γ and Δ), fitting a unique solution requires at least two independent tracer measurements.

For steady-state 1D flow, $G(t')$ takes an inverse-Gaussian form:

$$G(t') = \sqrt{\frac{\Gamma^3}{4\pi\Delta^2 t'^3}} \exp\left[-\frac{\Gamma(t' - \Gamma)^2}{4\Delta^2 t'}\right]. \quad (2)$$

We also compute t_{mode} , the most probable age in the distribution (Smith et al., 2011; Wefing et al., 2021):

$$t_{\text{mode}} = \frac{1}{\Gamma} \left(\sqrt{9\Delta^4 + \Gamma^4} - 3\Delta^2 \right). \quad (3)$$

We use the variable-geometry “Smith TTD” (Raimondi et al., 2024), where Δ and Γ both vary. The full Δ/Γ grid is mapped in tracer–tracer space, and each measurement is assigned to its nearest grid point. Points falling outside the gridded domain are excluded.

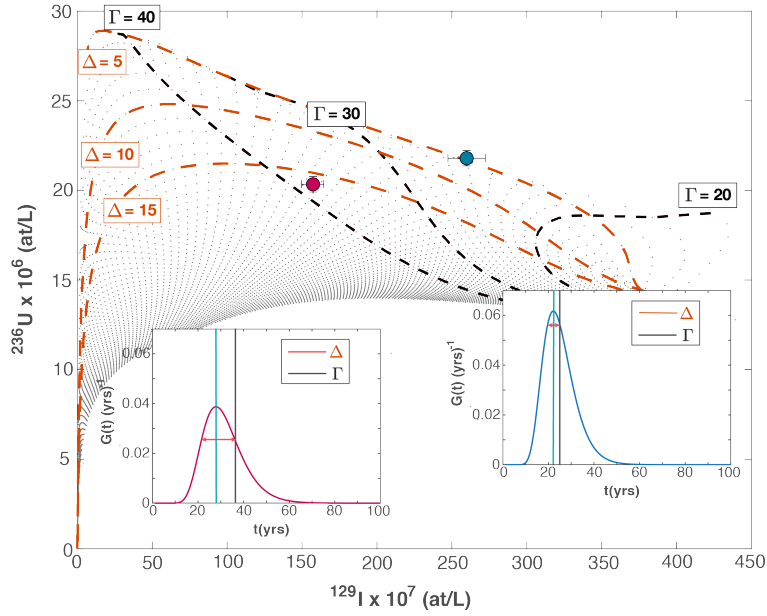


Figure 1: Example TTD grid, with the obtained PDF marked by colour. Isolines of Δ and Γ are marked. On the PDF, the orange horizontal line is Δ , the black vertical line is Γ , the bright blue line is t_{mode} .

5.2 Running the calculation

You can calculate TTDs using radionuclides and, traditionally, the gas tracers CFC-12 and SF₆. The workflow is:

1. Load input functions for each tracer on a 0.1 yr timestep.
2. Build a matrix of forward-modelled tracer concentrations by evaluating Eq. 1 for every (Γ, Δ) pair across your grid.
3. For each measurement, find the grid point that best matches the observed tracer concentrations (minimum residual in tracer–tracer space) to recover Γ , Δ , and t_{mode} .
4. Run 1000 Monte Carlo iterations to propagate measurement uncertainties into uncertainties on Γ and Δ .

You will need the following functions and scripts:

- **fun_inverse_gaussian.m** — evaluates Eq. 2.
- **fun_t_mode.m** — evaluates Eq. 3.
- **calc_matrix.m** — builds the matrix of ^{129}I and ^{236}U concentrations across the (Γ, Δ) grid (step 2).
- **calc_G_D.m** — main driver script; calls the matrix calculation and the grid search to return Γ and Δ for each sample.
- **fun_find_closest_D_G_MonteCarlo.m** — runs the grid search with Monte Carlo uncertainty propagation (steps 3–4).

6 C_{ant} from the TTD

Once you have Γ and Δ from the previous section, you can calculate the anthropogenic carbon concentration (C_{ant}) at each sample point. Following Waugh et al. (2006), we assume C_{ant} behaves as a passive tracer — that is, once it enters the ocean interior it is transported by the same circulation as any other dissolved substance, with no internal sources or sinks. Under this assumption, C_{ant} satisfies the same integral as Eq. 1 but without radioactive decay:

$$C_{\text{ant}}(x, t) = \int_0^\infty C_{\text{ant},0}(t - t') G(x, t') dt'. \quad (4)$$

Here $C_{\text{ant}}(x, t)$ is the unknown interior C_{ant} concentration (it cannot be measured directly), $C_{\text{ant},0}(t)$ is the reconstructed surface history (see next subsection), and $G(x, t')$ is the age spectrum from Eq. 2, evaluated using the Γ and Δ already solved for.

6.1 Required functions and toolboxes

- **CO2SYS.m** (v1.1) (Lewis & Wallace, 1998; Van Heuven et al., 2011) — computes the full carbonate system from any two of DIC, alkalinity, pH, and $p\text{CO}_2$.
- **co2atmhist.m** — provides atmospheric CO_2 history for constructing $C_{\text{ant},0}(t)$.
- **fun_Cant_Smith_TTD.m** — main driver; takes Γ , Δ , salinity, potential temperature, and sample year and returns C_{ant} for each sample via Eq. 4.
- TEOS-10 toolbox (for salinity and potential temperature inputs).

Example call:

```
CO2_ant = fun_Cant_Smith_TTD(D, D.Salinity, D.Potential_Temp, D.SampleYear, 1)
```

Make sure potential temperature is available for all samples - section 4.3.

6.2 Reconstructing the surface history $C_{\text{ant},0}(t)$

The surface history is reconstructed in three steps, following Raimondi et al. (2024):

1. Take mid-year atmospheric CO_2 concentrations for the Northern Hemisphere and subtract the pre-industrial baseline of 280 ppm. This isolates the anthropogenic component of atmospheric CO_2 , which is converted to a partial pressure perturbation pC_{ant} .
2. Estimate preformed total alkalinity from salinity using the relationship of Brewer et al. (1986):

$$A_T = 547.05 + (50.56 \times S) \quad [\mu\text{mol kg}^{-1}] \quad (5)$$

3. Input pC_{ant} , A_T , and surface temperature and salinity to CO2SYS to compute the anthropogenic DIC perturbation. Carbonate dissociation constants (K1, K2) are from Roy et al. (1993).

The resulting time series $C_{\text{ant},0}(t)$ is the input function for Eq. 4. This method assumes constant air-sea disequilibrium (i.e. the offset between surface ocean and atmospheric $p\text{CO}_2$ is assumed fixed over time) and full equilibration of anthropogenic CO_2 with the surface mixed layer.

7 Column and regional inventories of C_{ant}

Column inventories integrate C_{ant} vertically through a representative profile:

$$I_{C_{\text{ant}}} = \int_0^{z_{\text{max}}} C_{\text{ant}}(z) \rho(z) dz \quad (6)$$

where $I_{C_{\text{ant}}}$ has units of mol m^{-2} , $C_{\text{ant}}(z)$ is in mol kg^{-1} , $\rho(z)$ is the in-situ density in kg m^{-3} , and z_{max} is the integration depth (typically the deepest sample in the profile, or a fixed isopycnal).

Regional inventories are obtained by multiplying the column inventory by the area of the region of interest. Choose the area based on the domain used in the study; typical units for regional inventories are Pg C.

There is an example code available **calc_carbon_basin_inventory.m**.

8 Key Papers

To save you time, a complete BibLaTeX library is also available on GitHub with about 350 references (it's Arctic-focused). It includes key papers on AMS, iodine, uranium, the gas tracers, argon-39, carbon-14, anthropogenic carbon, Arctic circulation, and biogeochemistry.

References

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